Final paper

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Introduction

Restoration efforts at the North Campus Open Space (NCOS) are trying to increase species richness across major habitats, where key species are driving changes in composition, habitat stability, and biodiversity. 10 native and non-native plants were selected for this study based on high observations, presence during all monitoring years (2021-2025), and frequent mention in the 2024 monitoring report and native plant catalog. This study specifically looks at how species responded to a prescribed burn that happened in Fall 2023 as well as the effects of a year with heavier rainfall in 2024. Therefore, the main questions for this study include, **"how has percent cover of key native species and overall native species richness changed through time in grasslands compared to non-native species?"** and **"What impact does the 2023 fire and 2024 water year have on species richness and cover for both native and non-native grassland species?"**

Our native species of interest are:

`Deinandra fasciculata`, `Distichlis spicata`, `Erigeron canadensis`,
`Stipa pulchra`,
`Symphyotrichum subulatum`,

And our non-native species of interest are:

`Bromus hordeaceus`, `Festuca myuros`,
`Festuca perennis`,
`Medicago polymorpha`, `Polypogon monspeliensis`.

This study will pay close attention to Stipa pulchra — California's state grass — as well as Festuca perennis, a non-native grass, as these species are greatly abundant across all eight grassland transects. Lastly, Deinandra fasciculata is a species endemic to the Santa Barbara region and was stated to have "responded well to the fire", which might indicate that native species perform better after a prescribed burn (Rickard 2024). The monitoring report also stated that S. pulchra was predicted to respond well to the fires. Other factors besides native/non-native status will be taken into account in this study, such as perennial versus annual grasses. Monitoring plant cover and richness over time can best inform restoration experts how certain intentional or unintentional events affect overall restoration progress. A study conducted on a grassland habitat found that species show positive responses to water availability extremes as net primary productivity (NPP) increased during wet years. Additionally, functional diversity was considered an indicator of healthier ecosystems and more stable NPP, which can be informed by *structural* diversity, such as total cover and richness (Alexander et al. 2023). This is significant in the case of NCOS, as increasing native species abundance and diversity can enhance ecosystem health and resilience.

Methods

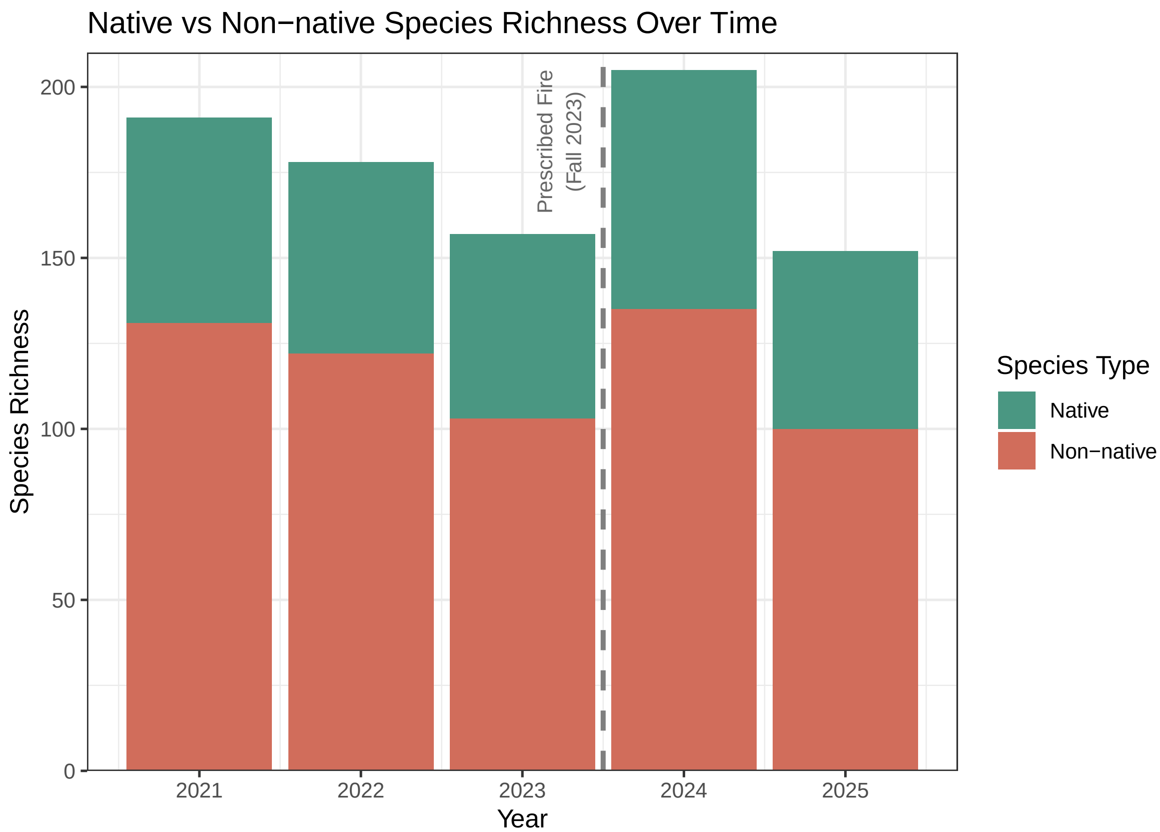
Data Collection and Study Area

To answer our research question, we used a vegetation dataset sourced from the Cheadle Center for Biodiversity and Ecological Restoration (CCBER), and provided to us by the ENVS 193DD course. This dataset spans 5 years (2021-2025) of vegetation data from across several habitats in North Campus Open Space (NCOS) including grasslands, mosaic habitats, wetlands, open water, woodlands, and shrublands. We chose to focus on native perennial grasslands, which encompasses measurements from 8 different transects across the 16.8-acre mesa. The method of data collection used for vegetation surveys is the Quadrat Transects (QT) method. This method involves the use of 11 one-square-meter quadrats placed by a 30-meter transect, with quadrats alternating between the left and right side of the transect line every 3 meters. "The quadrats are subdivided into 100 ten-centimeter squares and the percent cover is estimated [by human monitors] for each species in the quadrat," (NCOS Restoration Project Monitoring Report).

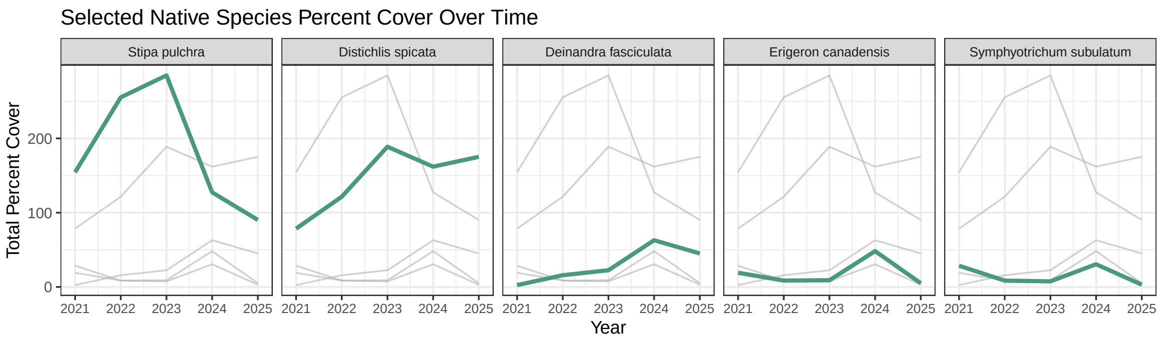
Analytical Approach

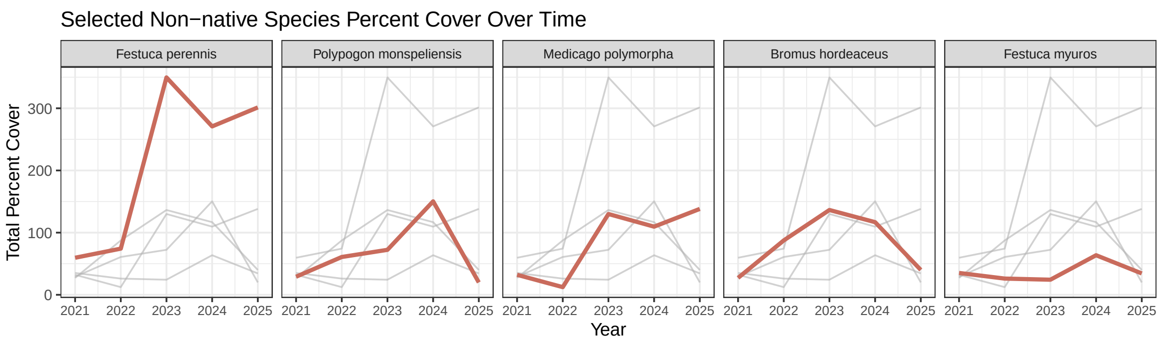
We will not be performing a formal statistical analysis for our project because we are primarily exploring a correlative relationship rather than predicting trends for species cover and richness. However, we will be identifying major events that might influence a directional response in vegetation percent cover and species richness, and examining whether a response was captured in our dataset. Major events include the prescribed burn in Fall 2023 because of its effects as an intermediate disturbance event, as well as wet (2023, 2024) and dry (2021, 2022) water years because water is a limiting resource for key native grass species (S. pulchra). These factors are potential influences on how percent cover and species richness change over time. This informal analysis will be conducted by examining the visual components of our figures. These visual components include changes in size or direction of geometries representing key variables, as well as geometries marking important contextual events.

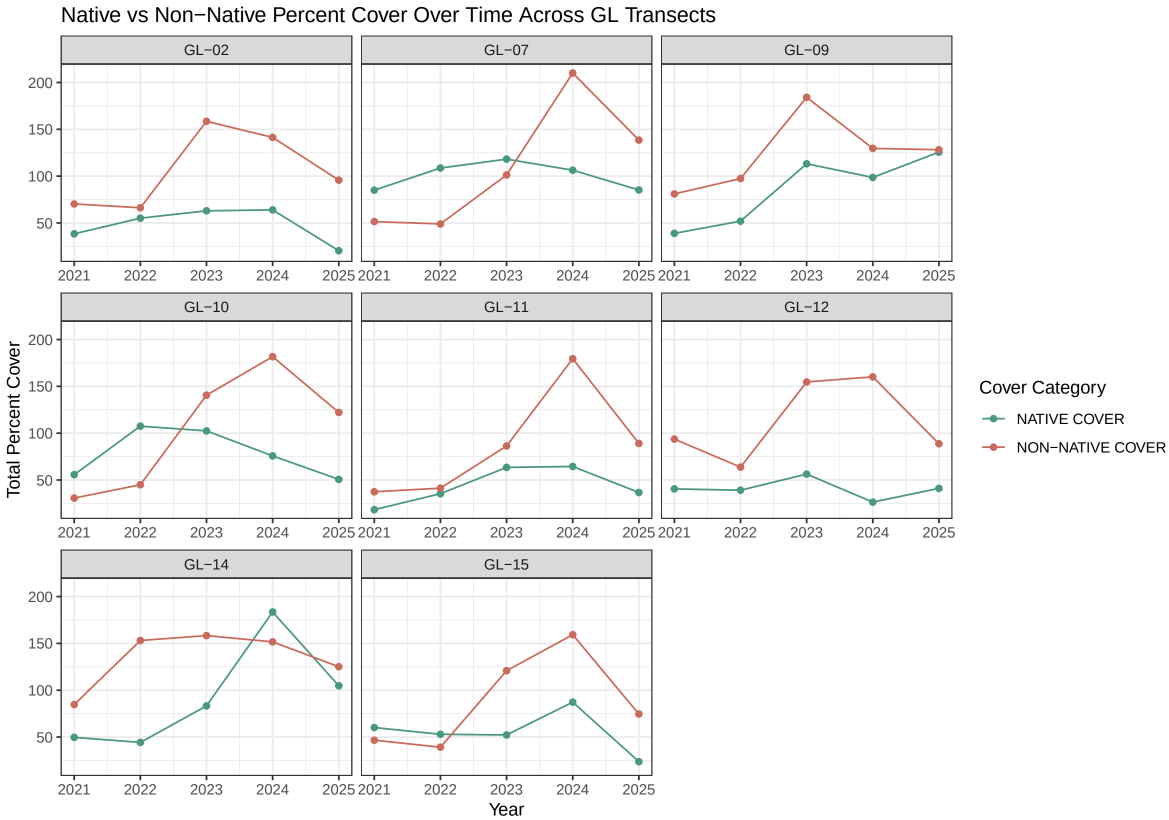
Results



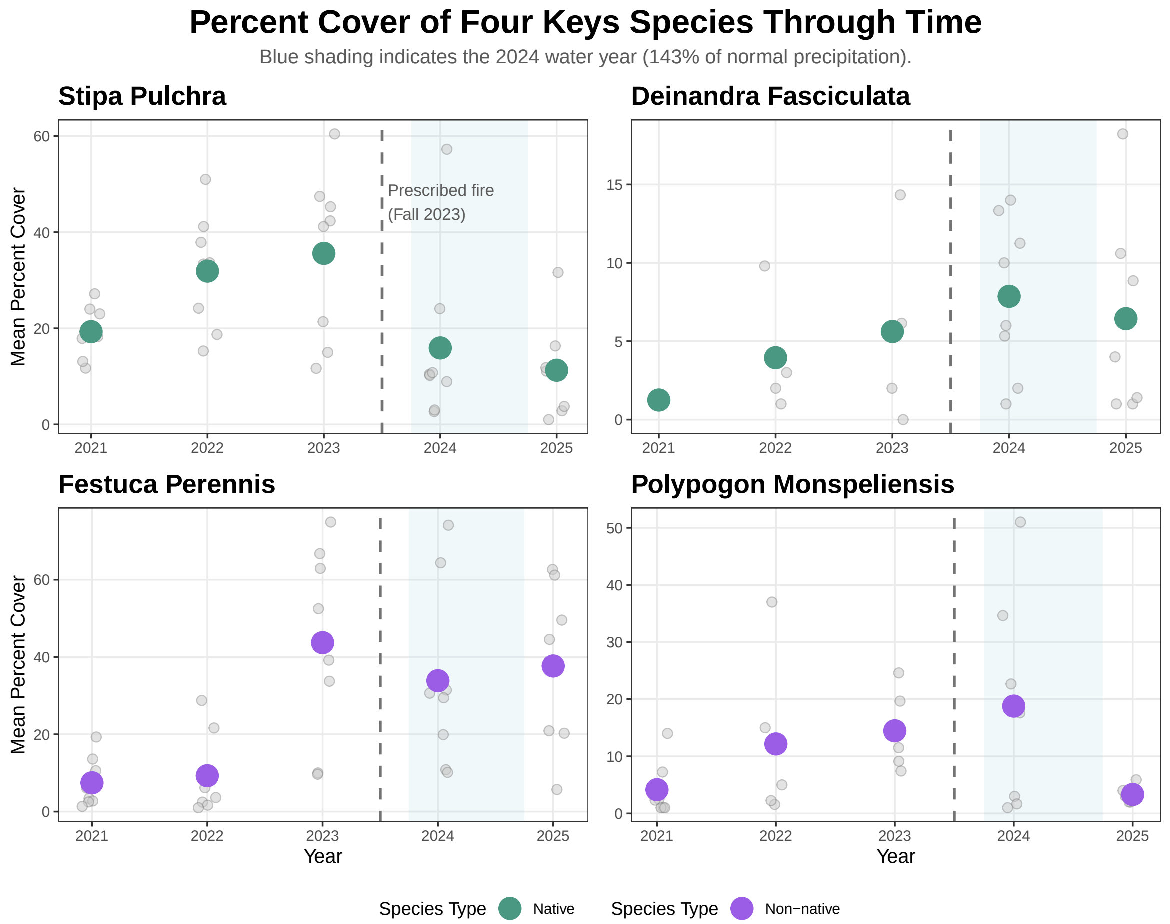
Native vs Non−native Species Richness Over Time
Prescribed Fire
(Fall 2023)
200
150
100
50
0
Species Richness
2021
2022
2023
2024
2025
Year
Species Type
Native
Non−native


Selected Native Species Percent Cover Over Time
Stipa pulchra
Distichlis spicata
Deinandra fasciculata
Erigeron canadensis
Symphyotrichum subulatum
Total Percent Cover
0
100
200
2021
2022
2023
2024
2025
Year


Selected Non−native Species Percent Cover Over Time
Festuca perennis
Polypogon monspeliensis
Medicago polymorpha
Bromus hordeaceus
Festuca myuros
Total Percent Cover
0
100
200
300
2021
2022
2023
2024
2025
Year



Native vs Non−Native Percent Cover Over Time Across GL Transects
GL−02
GL−07
GL−09
GL−10
GL−11
GL−12
GL−14
GL−15
Total Percent Cover
Year
200
150
100
50
2021
2022
2023
2024
2025
Cover Category
NATIVE COVER
NON−NATIVE COVER



Percent Cover and Species Richness Trends

Overall, our results show that both native and non-native species richness increased following the prescribed burn of fall 2023 and during the 2024 water year, with non-native richness having the largest increase. Concerning percent cover, we saw native percent cover increase in transects 14 and 15 after the burn, which are the closest transects to vernal pools 1 and 2. Non-native percent cover after the prescribed burn increased in transects 7, 10, 11, and 15, with non-native cover generally remaining relatively high across most transects during the study period. Spatially, as mentioned, we found that native species percent cover increased most strongly near vernal pools 1 and 2, which is significant because these vernal pools also experienced the greatest water depths during the 2024 water year. Lastly, we found that non-native percent cover and overall species richness increased most strongly in the southeastern grassland transects.

Focal Species Trends

We also found several meaningful patterns concerning our focal species. The percent cover of

Stipa pulchra, a key NCOS restoration target species, increased from 2021 to 2023; however, it declined after the burn and the 2024 water year, overall decreasing over time. The percent cover of Deinandra fasciculata, a species endemic to Santa Barbara, remained relatively low throughout the study period; however, percent cover increased slightly following the prescribed burn and during the 2024 water year. Concerning non-native focal species, we found that Festuca perennis, another restoration target, decreased slightly after the burn; however, it rebounded in 2025 and now has one of the highest percent covers out of any species. Polypogon monspeliensis did not seem to react to the burn as it increased slightly in 2024; however, it may have been outcompeted by other species during the 2024 water year, as it declined substantially in 2025 and now has a very small percent cover.

Discussion

Connection to Existing Publications

Concerning Stipa pulchra, our results supported other relevant studies that also determined that prescribed fires had little positive effect on Stipa pulchra, a key NCOS restoration target (Keeley et al. 2023). Its substantial decline in percent cover after the prescribed burn of fall 2023 is likely due to being outcompeted by invasive opportunistic annuals who are establishing faster following disturbances (Gornish and Shaw 2017; Hamilton, Holzapfel, and Mahall 1999). As said in the monitoring report, it will likely take longer observation to determine whether declines in Stipa pulchra represent short-term responses to disturbance events or long-term trends. While we did see Festuca perennis decline slightly after the fire, it did rebound in 2025 and has increased over time; however, the rate of its increase has slowed after multiple disturbance events. Other grassland studies also found that Festuca perennis did increase after burns; however, it did have negative trends following large precipitation events (Keeley et al. 2023). This could possibly be the reason why its rate of increase in percent cover has slowed after 2023 (because of the heavy 2024 water year). Our results also concur with the Gornish & Shaw study, which concluded that Deinandra fasciculata responds well to disturbances, as we have seen an overall increase in its relatively low percent cover over time, especially after the prescribed burn. Our results concerning how native and non-native species reacted to disturbance events such as the fall 2023 prescribed burn and 2024 water year followed similarly to existing studies. Among our key species, we consistently found that non-native annuals were outcompeting native perennials, as trends between the two were often inverse (Corbin and D'Antonio 2010; Gornish and Shaw 2017). This is due to annual species with shallow root systems often being able to establish quicker after disturbance events. Even among our key perennial and annual species, without distinguishing between native and non-native species, we found generally annual species were outperforming perennials; for example, even though we saw Stipa pulchra decline, Deinandra fasciculata, an annual native, has increased over time. One notable exception, however, is Distichlis spicata, which is a native perennial that has been thriving at NCOS. This is likely due to it being a very salt-, drought-, and stress-tolerant species that has adapted to many California habitats (Pessarakli and Marcum 2012).

Relevance to Other Ecosystems

Our results allow us to make inferences about California grassland ecosystems. The patterns exhibited as NCOS have been well-documented among several California grassland studies; disturbance events such as fires, sheep grazing, and wet water years often favor fast-growing annuals over slow-establishing perennial grasses. The decline of Stipa pulchra and the increase of fast-growing annuals reflect the overall pattern of California grasslands in which annuals are outcompeting native perennials. One new pattern we found from our study is that native percent cover generally increased in transects located near vernal pools 1 and 2, which are the deepest vernal pools at NCOS that had the highest water input during the 2024 water year. This suggests that greater water availability may be benefiting native species establishment and that ecosystems with deeper vernal pools may be more beneficial to native perennial species because deeper root systems may have greater access to water.

Limitations

There are certain limitations to consider that could affect the results of our study. First, the vegetation data is a survey that is taken annually during July, rather than multiple times throughout the year. Therefore, seasonal fluctuations were not able to be visualized in the analysis. The nature of restoration monitoring involves subjectivity during observations when cover is estimated visually from each transect. This means that measured cover can be easily over- or underestimated. Additionally, the prescribed fire only occurred once during the time that the vegetation data set was recorded. While we did see evidence of the fire having a more equitable effect on native and non-native species alike, more events in the future should be taken into consideration to reach an interpretation with higher confidence. It is hard to establish causation considering that the fire occurred in the Fall of 2023 and vegetation measurements were not recorded until July of 2024. Other factors like annuals versus perennials, proximity to water abundance (vernal pools), and proximity to trails with increased human disturbances were not captured explicitly within the vegetation data set but are acknowledged in this study.

References

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